Cutoff: terrain-gated SAR flood/debris mapping with evidence-graded settlement isolation

Multimodal AI Hackathon 2026 · Track B "Mapping Flood Damage from Space" · prepared 10 Oct 2026

Labels used throughout: **[V]** verified by me against a primary source in this session · **[S]** appears only in your supplied documents, not re-verified · **[D]** design decision · **[H]** hypothesis to test · **[?]** open fact that must be checked.

1. Executive decision

Build "Cutoff": one pipeline that takes an area + flood date and runs raw Sentinel-1 (same-track pre/post) → a trained flood-segmentation model plus a terrain-gated change detector → OSM exposure → a scenario-based road-graph analysis of which settlements lose road access to the nearest town/hospital → a dashboard and a template-generated one-page situation report. It is evaluated three ways: on Kuro Siwo's official held-out events (stratified by terrain slope), on the Trishuli event against EMSR927 with **zero tuning on EMSR927**, and with a sensitivity study of the cut-off result to closure assumptions and OSM completeness.

AI component chosen: flood segmentation model (25% of the score, and the PS explicitly asks to show performance on unseen Himalayan scenes). The situation report is generated from a JSON of computed numbers by a template, so no language model can invent a number; the Nepali copilot is a stretch goal, not a requirement under this option.

Three decisions that differ from your supplied research, each with a verified reason:

1. **Training data = Kuro Siwo only (Sen1Floods11 optional, post-only baseline).** GEOID-Flood [V] and MMFlood [V] are built from Copernicus EMS delineations, are not on the PS dataset list, and GEOID-Flood spans 2016–2026, so it may contain EMSR927 itself. The PS says EMS products may only be used to check results, and breaking this rule means disqualification. The same logic rules out IBM ImpactMesh (EMS-derived labels, per your repo report [S]) and WorldFloods.
2. **No Swin/dual-stream transformer, no conformal stack in the first build.** There are no debris labels, Kuro Siwo's own paper reports DEM/slope inputs give only marginal gains [V], and the budget is 15 days. A U-Net, terrain gating and temperature scaling are

enough to defend.

3. **"Himalayan unseen scenes" is proven by Trishuli (never trained on) plus slope-stratified Kuro Siwo test events**, because no labelled Himalayan flood training data was identified in any supplied document.

Top three risks: live preprocessing latency on judge-chosen areas; no Himalayan labels beyond EMSR927; incomplete pre-event OSM in remote valleys. Mitigations are in section 16 of this document.

2. Official PS and track compliance matrix

Source key: **PS** = Track_B.pdf / Space_Track.pdf (identical content).

#	Requirement	Source	M/O	Our implementation	Verification
R1	System takes an area and a flood date	PS Requirements	M	CLI and dashboard accept AOI polygon/bbox + event date; nothing hard-coded to Trishuli	Run on ≥2 non-Trishuli AOIs
R2	Where did the flood hit: S1 before/after, S2 when clear	PS Q1	M	Same-track S1 change + trained model; S2 only as corroboration where SCL is clear	Overlap with EMSR927; S2 coverage flag
R3	What was damaged: OSM buildings, roads, bridges, estimate which were hit	PS Q2	M	Hazard ∩ OSM; reported as **exposed**, never "destroyed"	Recall of EMS-damaged features (check-only)
R4	Who is cut off: no road link to nearest town/hospital	PS Q3	M	NetworkX graph, three closure scenarios, four status classes	Synthetic-graph tests; EMS-closure replay
R5	AI component: choose one	PS AI component	M	Flood segmentation model	Section 11 protocol
R6	Show performance on unseen Himalayan scenes	PS AI option 1	M	Trishuli (no training, no tuning) + slope strata	Metrics table in report
R7	Bonus: trace flood path downstream, list settlements	PS Bonus	O	pyflwdir flow path + HAND corridor + OSM places	Path follows Lhende→Bhote Koshi→Trishuli

#	Requirement	Source	M/O	Our implementation	Verification
R8	Case study vs EMSR927	PS Case study	M	`eval/emsr927.py`, check-only	Agreement + disagreement taxonomy
R9	Map dashboard + one-page system-generated situation report	PS Output	M	Streamlit + Jinja→PDF	Demo, usability pass
R10	GitHub repo with README on how to run	PS Deliverables	M	`README.md`, `make` targets, pinned env	Fresh-clone run by a teammate
R11	Report ≤6 pages incl. limitations	PS Deliverables	M	`docs/report/`, limitations written from section 11 failure data	Page count
R12	3-minute demo video	PS Deliverables	M	Script in section 17	Timed run
R13	Attributions (3 exact strings) + training-dataset citations	PS Attribution	M	`ATTRIBUTION.md`; footer on dashboard, report and sitrep; cite Bountos et al. 2024	Grep check in CI
R14	Allowed inputs only: S1, S2, Copernicus DEM, pre-event OSM, listed training sets	PS Data rules	M (disqualifying)	Config allow-list; `pipeline` never imports `eval`; test asserts OSM date < 2026-08-26	`tests/test_no_reference_leak.py`
R15	OSM snapshot before 26 Aug 2026 (latest available 27 Jul 2026 per PS)	PS Datasets	M	Default `2026-07-27`, asserted	Test
R16	Published modified OSM data stays ODbL	PS Datasets	M	Any exported OSM-derived GeoJSON carries ODbL notice	README
R17	Compare same orbit track only (same time of day, 12 days apart)	PS Tip	Strongly implied	Pair selector enforces relative orbit; cross-track only in a labelled ablation	Log + test
R18	Run end-to-end from raw data for any judge-chosen area/date, live	PS Data rules, Evaluation	M	Section 8 pipeline, caching, timing tests	Runtime test E7

#	Requirement	Source	M/O	Our implementation	Verification
R19	No victim images; educational prototype wording	PS Responsibility	M	Maps only; disclaimer on every output	Review
R20	Limitations: what the system cannot do	PS Limitations	M	e.g. no early warning given revisit time	Section 6 and report

Scoring weights from the PS: flood/damage mapping 30%, AI component 25%, cut-off analysis 20%, report/Q&A/honesty 15%, dashboard/report usability 10%. Effort allocation follows these.

3. Research audit

3.1 Source ledger

Source	Type / date	Verified by me?	What I used	Limits
Track_B.pdf, Space_Track.pdf	Official PS, 2026	**[V]** read in full	All rules and weights	Illustrative images only
Kuro Siwo GitHub README, github.com/Orion-AI-Lab/KuroSiwo	Repo page, fetched today	**[V]**	`main.py --method --backbone --dem --slope`; GRD + SLC; GeoTIFFs and webdatasets on Hugging Face; polygons repo; weights for FloodViT and SNUNet (Dropbox); SNAP preprocessing XML in `configs/`; MIT LICENSE file while README text says CC BY	Dataset download size not stated

Source	Type / date	Verified by me?	What I used	Limits
Bountos et al., NeurIPS 2024 D&B (arXiv 2311.12056)	Peer-reviewed	**[V]** abstract and tables via search	43 events; 67,490 labelled 224×224 patches at 10 m; 2 pre + 1 post S1 per event; DEM and slope; classes no-water / permanent water / flood; 10 test events; best models >80% F1 on flood; DEM/slope gives marginal gains; permanent water is the weakest class	I did not read the full paper; event geography not seen
CEMS EMSR927 page, mapping.emergency.copernicus.eu/news/flood-in-nepal-emsr927/	Official, 26 Aug 2026	**[V]**	Activation scope: flood extent + damage assessment; "reportedly" a GLOF	Vector package contents not seen **[?]**
EU Space "Image of the day" ×2	Official	**[V]**	Four AOIs; >240 buildings destroyed and 32 damaged in the Syapru Besi view; Sentinel-2 scenes of 12 and 27 Aug near Betrawati/Gerkhu show wider sediment-filled channel	One AOI view only
Community catalog "Nepal Floods 2026" (gee-community-catalog.org)	Secondary, updated 31 Aug	**[V]**	Flood at ~02:52 UTC 26 Aug; ~100 km runout; casualty and damage figures explicitly provisional; Planet/Maxar imagery is CC BY-NC and mostly cloudy	Secondary source; not an allowed input
GEOID-Flood (arXiv 2608.02315)	Preprint 2026	**[V]** abstract + §3	CEMS-derived, 219 events, S1 RTC + S2 + GLO-30, CC BY 4.0; fusion best for transient flooding	Not on PS list; leakage risk
MMFlood (IEEE Access 2022)	Peer-reviewed	**[V]** abstract	95 EMS events, S1 + DEM + hydrography	EMS-derived
ICIMOD "Flood data of Nepal, 11 Aug 2017" (doi 10.26066/RDS.33614)	Dataset, CC BY 4.0	**[V]** snippet	S1-derived flood extent over Nepal: candidate check-only scene	Likely lowland; format unseen
NDRRMA Sitrep #1, Rasuwa glacial flood, July 2025	Official	**[V]** snippet	A 2025 outburst flood on the same corridor: candidate second Himalayan scene	Reference labels unknown **[?]**

Source	Type / date	Verified by me?	What I used	Limits
ohsome API docs (docs.ohsome.org)	Official docs	**[V]** snippet	`time` and `filter` parameters; geometry endpoint	Response limits to test
Elicit review PDF + sources.xlsx	AI-assisted literature review	**[V]** read; spreadsheet = 16 rows matching the PDF's references	Gap list; 15-day scope recommendation; dual-pol, radiometric-aug and held-out-event advice	Abstract-level only (its own caveat); PDF says 19 papers, sheet has 16
Doc A: GitHub repo investigation (pasted agent log)	AI agent report	**[S]**	Licence notes (GPL-3.0 reference-only; Sen1Floods11 no licence)	Rankings and "7/7" scores not reproducible
Doc B: deep-research briefing (pasted)	AI report	**[S]**	Reporting rule (exposure ≠ damage); closure-scenario structure; repo list	Numbers from Springer/ILO/UNOSAT not re-checked
Doc C: Himalayan_Flood_Mapping_AI_Assessment PDF	AI report	**[S]**	Physical failure modes; falsification-experiment idea	Several inconsistencies, see 3.2
Sen1Floods11 (Bonafilia, CVPRW 2020)	Peer-reviewed	**[S]** via Elicit	4,831 chips, 11 events, mostly auto-generated labels	Single-date S1

3.2 Contradictions and decisions

Claim in supplied material	Problem	Decision
Doc A: no repo covers more than 4 of 7 components, so the brief is "genuinely novel"	Doc B lists repos that already do S1 change detection + OSM road impact, a dashboard, and flood-to-healthcare routing **[S]**	Claim novelty only for engineering and evaluation
Doc A: fork IBM ImpactMesh	EMS-derived labels cannot be inputs; not on PS list; I did not verify it	Reject
Doc B: fine-tune on Sen1Floods11 and/or WorldFloods	WorldFloods not on PS list; Sen1Floods11 has one S1 date, so it cannot teach pre/post change	WorldFloods out; Sen1Floods11 optional post-only baseline

Claim in supplied material	Problem	Decision
Elicit/Doc B: use GEOID-Flood, UrbanSARFloods	EMS-derived (GEOID) or not on the list; leakage or disqualification risk	Related work only
Doc C: Kuro Siwo models "fail to detect debris" and DEM is essential	Plausible because Kuro Siwo has no debris class **[V]**, but unevidenced; paper says DEM is marginal **[V]**	Test it as H2/H3, do not assume
Doc C: dB thresholds (−4/+3/+8 dB), HAND >15 m vs <30 m vs >5 m, blank slope limit	Thresholds cited to a Chamoli water-quality paper, internally inconsistent	No hard-coded dB limits; scene-adaptive thresholds (section 8)
Doc C: three-fold plan training on Annapurna/Karnali/Chamoli basins	No labelled Himalayan flood data identified	Replace with Kuro Siwo events + Trishuli zero-tuning check
Doc C: isolated-village precision/recall against CEMS maps	I found no evidence CEMS publishes isolated villages **[?]**	Validate closed edges/bridges against EMS damage layers; replay routing
Doc C: Swin dual-stream, temporal attention, conformal, super-resolution	Not deliverable in 15 days; no debris labels	Simple U-Net, temperature scaling
Doc C: ascending Track 56 vs descending Track 63	Unverifiable here (no CDSE query possible)	Tracks discovered at run time
Event cause and casualty counts differ (PS: glacier collapse; CEMS: "reportedly" GLOF; catalog: unresolved; death tolls rising)	Sources disagree and are provisional	Never print casualty figures; use "reported" wording

3.3 What I could not verify

Sentinel-1 acquisitions, relative orbits and download latency for the corridor; contents of Kuro Siwo's `grd_preprocessing.xml` (units, DEM used, speckle filter); Kuro Siwo event geography and download size; input format of the released weights; EMSR927 vector

layers and legend; the code of repos named in Docs A/B (ImpactMesh, Prithvi, healthcare-access-idai, SAR-Flood-Detection); 2026 Sentinel Hub quotas on CDSE; pyflwdir API details. Each is a Milestone 2–3 check.

4. Prior-art comparison matrix

System	Approach / data	Reported evidence	Limits	What Cutoff adds	Already done elsewhere?
Kuro Siwo + BlackBench **[V]**	Multi-temporal GRD/SLC, DEM+slope, U-Net/SNUNet/FloodViT-type models	>80% F1 (flood) on 10 test events (abstract)	Water classes only; DEM marginal; test geography not slope-stratified in what I saw	Slope-stratified reporting, terrain gating, Himalayan check	Training/benchmark yes; strata and gating no evidence
Sen1Floods11 **[S]**	4,831 chips, hand-labelled + weak labels	Basis of many papers	Weak labels, single date	Not used for training	n/a
GEOID-Flood, MMFlood **[V]**	EMS-derived masks with S1/DEM(/S2)	Fusion best for transient water	Barred by PS rules	Related work only	n/a
SAR-Flood-Detection (JakubNiedzwiedz) **[S]**	Refined Lee, change detection, JRC water removal, slope filter, OSM flooded roads	Workflow, no benchmark	Heuristic; shadow/wet debris confusion	Learned model + held-out evaluation + graph isolation	Road-flooding yes; benchmarked evaluation not seen
sentinel1-sar-flood-mapping (AliForghani) **[S]**	ASF search, co-registration, change detection, Streamlit	Workflow	No routing/OSM impact seen	Same-track enforcement + routing	Dashboard yes
healthcare-access-idai / Petricola 2022 **[S]**	S1 flood + OSM + OpenRouteService isochrones	Raster vs network methods differ by ~300,000 people (≈800k vs 500k)	Flood is a mask; no bridge logic or uncertainty	Closure-scenario range, bridge rule, status "unknown" for OSM gaps	Cut-off concept yes
DISruptionMap (arXiv 2410.05286) **[S]**	OSM + depth threshold (27 cm)	Preprint	Depth rule unsuited to debris/scour	Hazard probability, not depth	Accessibility yes
Baige outburst-flood study (Yang 2021) **[S]**	S1/S2/DEM + RF	OA 92.3%, κ 0.89, single event	Single case	Multi-event evaluation	Mountain fusion yes
CEMS EMSR927 **[V]**	Analyst VHR product	Reference	Not reproducible	Used as check only	Reference

No row contains our exact combination, but I cannot prove absence: I searched, I did not exhaust.

5. Evidence-backed gap

Candidate gap	PS weight	Evidence it exists	Feasible in 15 days	Verdict
G1 Mountain false alarms (shadow/layover, steep slopes) and no Himalayan held-out proof	30% + 25%	PS asks for it; Elicit gap 3 (mountain debris studies are case studies) **[S]**; radar-geometry physics (textbook)	Yes	**Chosen**
G2 Hazard-to-passability-to-isolation link with explicit assumptions	20%	Elicit gap 4; Petricola raster-vs-network difference **[S]**	Yes	**Chosen**
G3 Debris vs water separation	Part of Q1	No debris class in Kuro Siwo **[V]**; opposite backscatter signs suggested **[S]**	Only rule-based, no labels	Secondary, flagged hypothesis
G4 Calibrated uncertainty	Part of honesty score	Elicit gap 5 **[S]**	Light version (temperature scaling)	Included, light
G5 Grounded reports	15% / 10%	Elicit: not shown to improve outcomes **[S]**	Easy (template)	Deliverable, not a contribution
G6 Foundation-model fine-tuning	none	Unproven for debris	No	Rejected

6. Final project definition and defensible novelty

Title: Cutoff: terrain-gated, same-track SAR flood/debris mapping with evidence-graded settlement isolation.

Pitch: Rescuers need to know where the valley floor changed and which villages lost their road. Cutoff gives that answer from free satellite data for any valley and date, says how sure it is, and shows which assumptions would change the answer.

Users: disaster responders and analysts (educational prototype, not operational).

What existing systems fail to address (supported part): flood extent, flooded roads and healthcare isolation each exist separately **[S]**; I found no single open workflow that couples same-track change detection, terrain gating, a trained model with slope-stratified held-out evaluation, and closure-scenario isolation analysis.

Exact contribution

1. *Engineering:* automated same-track pair selection plus terrain gating (layover/shadow, slope, HAND) feeding a trained model, running from raw data for any AOI/date.
2. *Evaluation:* slope-stratified held-out metrics, a zero-tuning Trishuli check, and a cut-off sensitivity study (closure assumption, OSM completeness).
3. *Application:* Himalayan valley adaptation with a rule-based "disturbance" layer for what the water model cannot see.

Not claimed: research novelty (no new architecture or loss), operational readiness, validated damage or debris classification, or uniqueness.

Evidence needed to defend each claim: (1) a run log on two non-Trishuli AOIs; (2) tables E1–E2, E4 in section 11; (3) table E6 in section 11.

Limitations to state: revisit gaps (cannot warn of a sudden collapse); 10 m pixels versus narrow channels; imagery may show the flood after it peaked; wet smooth mud can look like water; OSM gaps create false "cut-off" results; Kuro Siwo labels water, not debris; EMSR927 is an analyst product, not ground truth.

7. Research questions and hypotheses

All are **[H]** until measured.

ID	Hypothesis	Test	Falsified if
H1	Pre-mean + post + difference inputs beat post-only on held-out events	E3	Flood IoU gain <2 pts on test events

ID	Hypothesis	Test	Falsified if
H2	Terrain gating cuts false-positive area on slope >15° by ≥30% with ≤5 pts recall loss	E2/E3	Either bound missed
H3	Cross-track pairs give more false positives on steep stable terrain than same-track pairs	E5	No difference beyond noise
H4	Fixed Kuro Siwo-tuned thresholds transfer to Trishuli with F1 within 15 pts of the in-domain test	E1 vs E4	Larger drop (reported honestly)
H5	The cut-off set changes materially across closure scenarios and under 10–20% OSM edge dropout	E6	Set stable (then say so)
H6	Backscatter increase (roughness) and decrease (water/smooth mud) separate two hazard types along the valley floor	E4 visual + EMS agreement per class	No class-wise agreement

Thresholds in the table are targets I set before experiments, not promises.

8. End-to-end architecture

```
Input: AOI polygon + event date (+ OSM date, default 2026-07-27)
  |
[1 DISCOVER]  CDSE STAC/OData -> same-track manifest: 1 post + up to 3 pre at
-12/-24/-36 d
  |
[2 PREPROCESS] S1: SNAP GPT (Kuro Siwo graph) -> dB VV/VH, shadow mask  |
DEM: GLO-30 -> slope, HAND, flow-dir
  |            S2 (optional): L2A + SCL -> pre composite, post, clear mask
  |
[3 HAZARD]    A: U-Net(pre-mean, post, difference, slope) -> p(flood),
p(permanent water)
  |           B: rule-based disturbance: |delta| > k * sigma_stable inside
HAND corridor
  |           C: gating (shadow/layover, slope, HAND) + optional S2
corroboration -> class + confidence
  |
[4 OSM]       ohsome snapshot -> buildings, roads (+bridge tag), places,
hospitals
[5 EXPOSURE]  hazard ∩ OSM -> exposed buildings, road km, bridges (by
confidence tier)
[6 NETWORK]   graph + 3 closure scenarios -> connected / degraded / cut-off /
unknown
[7 BONUS]     flow path from a point -> HAND corridor -> settlements along it
[8 OUTPUT]    results.json (single source of truth) -> Streamlit dashboard +
one-page sitrep PDF

Separate, check-only: eval/ reads reference/EMSR927 -> metrics (never
imported by stages 1-8)
```

Data flow, with schemas

#	Stage	Input	Output
1	Pair selection	AOI, date, STAC/OData	`pairs.json`: product IDs, relative orbit, direction, datetimes, coverage fraction
2	S1 preprocessing	GRD zips, DEM	`s1_{date}.tif` (2 bands VV/VH, dB, 10 m, UTM 45N for Trishuli), `shadow.tif`

#	Stage	Input	Output
3	Terrain	GLO-30 tiles	`dem.tif`, `slope.tif`, `hand.tif`, `flowdir.tif`
4	Hazard	stack above	`p_flood.tif` (float), `hazard.gpkg` (polygons: class, confidence tier, reason flags)
5	OSM	AOI, date	`osm_{layer}.gpkg`
6	Exposure	hazard + OSM	`exposure.gpkg`, counts in JSON
7	Network	roads, places, hospitals, hazard	`edges_status.gpkg`, `settlements.gpkg` (status per scenario)
8	Report	`results.json`	`dashboard`, `sitrep.pdf`

Role of each component: SNAP GPT does radiometric/terrain processing; pyflwdir derives HAND and flow paths; the U-Net supplies water probability; the rule-based detector covers what Kuro Siwo cannot label; NetworkX does routing; Streamlit and Jinja only read artifacts.

Core contribution vs the rest: stages 3 and 6 are the contribution; stages 1–2, 4–5 and 8 are supporting engineering.

Scene-adaptive thresholds [D]. For the disturbance layer, estimate sigma_stable as the robust spread (MAD) of the dB difference over pixels that cannot be flooded (HAND >100 m), per slope/aspect bin; flag |difference| > k·sigma_stable (k=3, set before looking at EMSR927) and keep connected areas above a minimum size. The positive and negative signs are reported separately. This avoids borrowing dB limits that I could not source.

Gating [D]. Mask layover/shadow from DEM + look geometry; downgrade (not delete) water-class pixels on steep slopes or high HAND to "possible"; physical limits are set from the training validation events, and the HAND search corridor is a conservative physical choice (start at 60 m, state it).

Failure cases and recovery

Failure	Detection	Recovery
No same-track post-event scene	Pair selector returns empty	Report "no valid observation" for the AOI; list next acquisition date; never fall back silently to cross-track
Cross-track only	Orbit mismatch	Run only as labelled ablation, flagged "not comparable"
Layover/shadow pixels	Shadow mask	Output "no valid observation" class, excluded from metrics
Persistent cloud (S2)	SCL clear fraction	S2 layer off; confidence not upgraded
Wet smooth mud ≈ water	Low-backscatter inside disturbance corridor	Tag "water or smooth mud (ambiguous)"
Missing bridge/trail in OSM	Settlement has no baseline route	Status "unknown", never "cut off"
Download or SNAP failure	Exit code, size check	Retry; cached scene fallback; fast-mode preprocessing

9. Data and labelling strategy

Dataset	Role	Modality / resolution	Labels / licence	Access	Known limits
Sentinel-1 GRD IW	Inputs (pre/post)	VV/VH, 10 m	Copernicus open licence **[V per PS]**	CDSE free account	12-day repeat, geometry limits
Sentinel-2 L2A	Optional corroboration	10–20 m	Open licence	CDSE	Monsoon cloud
Copernicus GLO-30 DEM	Slope, HAND, shadow	30 m	Attribution string required	AWS open data or CDSE **[?]**	Smooths narrow gorges
OSM via ohsome, 2026-07-27	Buildings, roads, bridges, places, hospitals	Vector	ODbL	`https://api.ohsome.org/v1`	Remote-area completeness

Dataset	Role	Modality / resolution	Labels / licence	Access	Known limits
Kuro Siwo (GRD)	Training + validation + official test	224×224 patches, 10 m, 2 pre + 1 post, DEM, slope	Manual labels, 3 classes; MIT file vs CC BY README **[V]**, cite Bountos 2024	GitHub + Hugging Face	Size to be checked **[?]**; labels are water, not debris
Sen1Floods11 (optional)	Post-only baseline only	S1 single date	No licence file; mirrors CC BY 4.0 **[S]**	GitHub	Weak labels, no pre-event scene
EMSR927	Check only	Vectors	Credit "European Union, Copernicus Emergency Management Service data"	mapping.emergency.copernicus.eu	Layers/legend unseen **[?]**
Other Himalayan checks (ICIMOD 2017, Rasuwa 2025)	Check only, if reference polygons exist	S1-derived extent	CC BY 4.0 (ICIMOD)	ICIMOD RDS	Availability **[?]**

What labels mean. Kuro Siwo marks flood water and permanent water. Our target for the PS question "where did the flood hit" is the union of model flood water and the rule-based disturbance layer, so the model's measured metrics describe the water part only; the report says so.

Leakage control

- Use Kuro Siwo's official train/test events; validation by whole event, never random patches.
- Drop any Kuro Siwo event within ~100 km of the Trishuli corridor from training.
- Normalisation statistics and thresholds come only from training/validation events.
- Nothing from Trishuli (imagery, orbit, labels) enters training or tuning; EMSR927 files live in `reference/` and are read only by `eval/`.

Immediately available: Kuro Siwo subset (after size check), GLO-30, ohsome. **Needs action:** CDSE account, scene downloads, organisers' answers on the open questions listed at the end of this document.

Download plan [D]: stratify a Kuro Siwo subset by event and flood fraction (target ~10–15k training patches); store float16 to fit a free Colab disk. Full float32 is far larger (8 channels × 224² × 4 B ≈ 1.6 MB per patch before compression, my arithmetic), so check

the real size first.

10. Model and baselines

Primary [D]: U-Net, ResNet-34 encoder, trained on Kuro Siwo with 7 input channels: pre-mean VV/VH, post VV/VH, post-minus-pre VV/VH, slope. Raw DEM is excluded (elevation range shifts across regions); HAND is used in post-processing only (it is meaningless inside a 2.2 km patch).

Why this one: the repo already supports `--method unet --backbone resnet18 --dem --slope` **[V]**; U-Net-family models are reproducible baselines and architecture differences were small in one benchmark **[S]**, so data representation and evaluation matter more than a bigger model.

Item	Choice [D]
Init	ImageNet encoder, first conv averaged to 7 channels; or scratch if that is worse on validation
Output	3 logits: no water / permanent water / flood
Loss	Class-weighted cross-entropy + Dice on flood class
Optimiser	AdamW, lr 3e-4, cosine decay, mixed precision
Batch / patch	32 / 224×224
Sampling	~50% patches containing flood
Augmentation	Radiometric (gain shift, speckle noise); no geometric by default, flips tested in ablation (geometric augmentation hurt in one benchmark **[S]**; flips alter look geometry)
Schedule	⩽30 epochs on the subset, checkpoint every epoch, early stop on validation-event flood IoU
Threshold	Flood threshold tuned on validation events only; also report PR curves
Calibration	Temperature scaling on validation events; report ECE

Item	Choice [D]
Reproducibility	Fixed seeds (1 minimum, 3 if time), pinned env, config snapshots
Compute	Free Colab T4 for training; CPU is enough for inference on small AOIs (my estimate, measure it)

Baselines

ID	Baseline	Purpose
B0	Change threshold (Otsu/robust) + permanent-water removal + HAND mask	Non-learned reference
B1	Released Kuro Siwo SNUNet weights, zero-shot (check input format first)	Published reference
B2	U-Net, post-only	Tests H1
B3	U-Net, pre+post+difference	Core model
P	B3 + slope input + gating (+ S2 corroboration)	Proposed

If P beats B3 on aggregate but loses on steep slopes, the conclusion is restricted accordingly (section 11).

11. Experimental and evaluation protocol

Metrics, in plain language

Metric	Meaning	Why included
IoU	Overlap of predicted and reference flood divided by their union	Standard, harsh on boundaries
Precision	Of pixels we call flood, how many truly are	False alarms waste rescue effort
Recall	Of true flood pixels, how many we found	Misses are the dangerous error

Metric	Meaning	Why included
F1 / Dice	Balance of precision and recall	Single comparable number
Stable-steep FPR	Share of pixels with HAND >100 m and slope >20° (cannot be flooded) flagged as flood	Direct measure of shadow/layover false alarms, needs no EMS data
Boundary F1 (30 m tolerance)	Edge agreement allowing for 10 m pixels vs VHR-derived reference	Fair to SAR resolution
Per-event and per-slope-stratum scores	Metrics split by event and by slope bin (0–5°, 5–15°, >15°)	Pooled averages hide terrain failures
ECE	Gap between stated confidence and observed accuracy	Tests whether the confidence map can be trusted
Closed-edge precision/recall	Our closed roads/bridges vs EMS-damaged transport features (25 m match)	Closest available check for cut-off logic
Runtime, peak RAM	Wall time and memory per AOI size	Live judging

Experiments

ID	Question	Data	Compared	Output
E1	How good is each model on held-out events?	Kuro Siwo official test events	B0, B1, B2, B3, P	Per-event table + pooled
E2	Does terrain change the result?	E1 patches binned by slope band	Same	Slope-stratified table

ID	Question	Data	Compared	Output
E3	Which component helps?	Validation + test events	Ablations: post-only, +pre, +difference, +slope, +gating, with/without flips, loss variants	Ablation table with seeds
E4	Does it transfer to the Himalaya, with no tuning?	Trishuli, EMSR927 as reference	P vs B0 vs B1	Pixel + boundary + object agreement, per AOI, plus a disagreement taxonomy (cloud, geometry, DEM, OSM, definition mismatch)
E5	Does same-track matter?	One Trishuli AOI, same-track vs a cross-track pre scene	P	Stable-steep FPR, both pairings
E6	How robust is the cut-off result?	OSM graph	(i) synthetic-graph unit tests; (ii) closure replay with EMS-damaged features; (iii) three scenarios; (iv) 20 random 10% and 20% road-edge dropouts	Table of status changes and flip rate
E7	Is it fast enough live?	10, 20, 40 km AOIs	Pipeline end to end	Time + RAM table
E8	Is confidence calibrated?	Validation events	Before/after temperature scaling	ECE overall and per slope bin

Targets set before training [D] (they may fail; we report either way)

- E1: P flood IoU at least 10 pts above B0; B3 within about 5 F1 pts of the published figure for comparable models on the same events **[verify the figure in the full paper]**.

- E2/H2: false-positive area on slope >15° reduced ≥30% with ≤5 pts recall loss.
- E4: no numeric target; the deliverable is an honest table. I will not promise an IoU.
- E7: ≤20 min for a 20×20 km AOI on a 16 GB laptop **[unmeasured target]**.
- Minimum to submit: E1, E4, E6(i) and E6(iii).

Interpretation rules

- Better pooled IoU but worse on slope >15° → claim only for flatter terrain and say so; do not average it away.
- Good Kuro Siwo score but poor Trishuli score → report domain shift, show the failure classes, do not retune on EMSR927.
- EMS disagreement is not automatically our error: the product is analyst-made from VHR optical imagery, may include different classes, and may be from a different date.
- Exposure counts are "exposed", damage is never inferred from a 10 m SAR cue alone.

12. Technology stack

Tool	Purpose	Alternatives considered	Free?	Verdict
Python 3.11, uv or conda	Everything	—	Yes	Required
PyTorch + segmentation_models_pytorch (or Kuro Siwo repo code)	U-Net training	Plain PyTorch; Kuro Siwo trainer	Yes	Reuse repo loader, own wrapper for extra channels
ESA SNAP + GPT	S1 radiometric/terrain processing matching Kuro Siwo	sarsen, pyroSAR, Sentinel Hub orthorectification, ASF HyP3	Yes	Primary for model consistency; heavy (RAM)
Sentinel Hub on CDSE (optional)	Fast-mode orthorectified S1	HyP3 queue	Free quota, limits **[?]**	Fallback; domain mismatch must be measured; ask organisers whether it counts as "raw"

Tool	Purpose	Alternatives considered	Free?	Verdict
pystac-client + requests	CDSE search/download	OData only	Free account	Required; verify endpoint
rasterio, rioxarray, geopandas, shapely, pyproj	Raster/vector work	GDAL CLI	Yes	Required
pyflwdir	HAND, flow direction, flow path	whitebox, pysheds, richdem	Yes	Chosen; verify API **[?]**
NetworkX	Graph and multi-source Dijkstra	igraph, OpenRouteService	Yes	Enough for ⩽ a few 100k edges
ohsome REST	OSM snapshot by date	Overpass attic date	Yes	Required by PS; Overpass only if organisers allow
Streamlit + leafmap/folium	Map dashboard	FastAPI + Leaflet/MapLibre	Yes	Chosen: Python-only, fastest to build
Jinja2 + WeasyPrint	One-page PDF from JSON	ReportLab	Yes	Template only; no LLM
pytest, GitHub Actions (optional)	Tests	—	Yes	Required for leakage test
Colab free (T4)	Training	Local GPU	Free with usage limits	Sufficient with a subset

Deliberately not used: Google Earth Engine (extra account/quota, hides preprocessing), foundation models, DVC, Docker orchestration, a separate backend service.

13. Repository structure

```
cutoff/
  README.md  ATTRIBUTION.md  LICENSE  environment.yml  Makefile
  configs/        default.yaml  paths.yaml  train_unet.yaml
closure_scenarios.yaml
  src/cutoff/
    data/   cdse.py  s1_pairs.py  s1_preprocess.py  dem.py  osm.py  s2.py
kuro_siwo.py
    geo/    terrain.py  change.py  gating.py  vectorize.py  exposure.py
network.py  flowpath.py
    ml/     dataset.py  model.py  train.py  infer.py  calibrate.py
    report/ schema.py  build_report.py  sitrep.html.j2
    pipeline.py  cli.py
  eval/           metrics.py  splits.py  baselines.py  ablations.py
emsr927.py  cutoff_eval.py
  app/            streamlit_app.py
  graphs/         s1_kuro_siwo.xml  (copied from Kuro Siwo configs, with
attribution)
  scripts/        download_kuro_siwo.sh  fetch_dem.sh  run_trishuli.sh
  tests/          test_pairs.py  test_network.py  test_gating.py
test_no_reference_leak.py
  notebooks/      colab_train.ipynb
  docs/           limitations.md  data_rules_audit.md  report/
  reference/      (git-ignored; EMSR927 files, read only by eval/)
  outputs/        (git-ignored; runs and checkpoints)
```

File	Responsibility	Inputs → outputs	Depends on	Ours or adapted
`data/s1_pairs.py`	Pick same-track pair; reject cross-track	AOI, date → `pairs.json`	`cdse.py`	Ours
`data/s1_preprocess.py`	Run SNAP graph on subset, write dB COGs + shadow mask	GRD, DEM → tifs	`graphs/*.xml`	Ours around Kuro Siwo graph
`data/dem.py`	Download/mosaic GLO-30	AOI → `dem.tif`	—	Ours
`data/osm.py`	ohsome calls (tiled), assert date < 2026-08-26	AOI, date → gpkg	—	Ours
`data/s2.py`	L2A + SCL clear mask, pre composite	AOI, dates → tifs	`cdse.py`	Ours (optional)
`data/kuro_siwo.py`	Event table, subset sampler, slope-strata index	repo/HF → index	—	Adapted
`geo/terrain.py`	Slope, aspect, HAND, flow-dir	`dem.tif` → rasters	pyflwdir	Ours

File	Responsibility	Inputs → outputs	Depends on	Ours or adapted
`geo/change.py`	Pre-mean, difference, sigma_stable, disturbance mask	S1 stack, HAND → masks	`terrain.py`	Ours
`geo/gating.py`	Shadow/layover/slope/HAND downgrade, S2 corroboration, confidence tier	prob + masks → classes	`change.py`	Ours
`geo/vectorize.py`	Raster→polygons with attributes	rasters → `hazard.gpkg`	rasterio	Ours
`geo/exposure.py`	Hazard ∩ buildings/roads/bridges by tier	gpkgs → counts	`vectorize.py`	Ours
`geo/network.py`	Graph build (split at shared vertices), scenarios, statuses	roads, places, hospitals, hazard → status gpkg	NetworkX	Ours
`geo/flowpath.py`	Bonus: path from point, settlements within corridor	DEM, point → list	`terrain.py`	Ours
`ml/dataset.py`	Kuro Siwo wrapper adding difference channels	index → tensors	`kuro_siwo.py`	Ours
`ml/model.py`, `train.py`, `infer.py`	U-Net, training loop, tiled inference	tensors → checkpoints, `p_flood.tif`	PyTorch	Ours using library encoders
`ml/calibrate.py`	Temperature scaling, ECE	val logits → T	—	Ours
`report/schema.py`	Pydantic schema of `results.json` (every reported number)	—	—	Ours
`report/build_report.py` + template	One-page sitrep with attribution footer	JSON → PDF	`schema.py`	Ours
`pipeline.py`, `cli.py`	Orchestrate stages 1–8	AOI, date → run folder	all above	Ours
`eval/emsr927.py`	Rasterise/compare EMSR927 layers	`reference/`, run outputs → tables	`metrics.py`	Ours, never imported by `pipeline`
`eval/cutoff_eval.py`	Closure replay, OSM dropout	graph + EMS closures → table	`network.py`	Ours
`tests/test_no_reference_leak.py`	Fails if `pipeline` imports `eval`/reads `reference/`, or OSM date ≥ 2026-08-26	source scan	—	Ours
`app/streamlit_app.py`	AOI/date form, run, layers, scenario toggle, evidence panel, sitrep download	run folder → UI	`pipeline.py`	Ours

Do not vendor the whole Kuro Siwo repo; import its loader code with attribution or copy the preprocessing XML and cite it.

14. Sequential implementation roadmap

Suggested split for up to four people: ML (M5–M8, M10); SAR/terrain (M3, M4, M9); OSM/network/exposure (M11, tests); app/report/video (M12, M15–M16). If solo, follow the critical path and drop the bonus and S2.

#	Milestone (days)	Tasks and files	Done when	Likely errors and fixes
M1	Requirement freeze (D1)	Fix section 2 matrix, assumptions, questions for organisers	Matrix signed off	Scope creep: anything not in the matrix waits
M2	Environment and repo (D1)	Repo, `environment.yml`, `Makefile`, CI optional, `ATTRIBUTION.md`	Fresh clone runs `make test`	GDAL/rasterio wheels: use conda-forge
M3	Data acquisition (D1–3)	CDSE account; `cdse.py`; download Trishuli same-track set; GLO-30; Kuro Siwo size check then subset; ohsome 2026-07-27	Files + checksums; Kuro Siwo event table	Slow/failed downloads: resume, pre-download overnight; ohsome limits: tile bbox
M4	S1 preprocessing (D2–4)	Read Kuro Siwo XML (units, DEM, filter); `s1_preprocess.py`; subset before heavy steps	Trishuli pre/post dB rasters align on stable terrain	SNAP out-of-memory: subset first, raise JVM heap; unit mismatch with Kuro Siwo: match its graph
M5	Dataset validation (D3–4)	`kuro_siwo.py`, plot patches, label sanity, slope-strata index	20 patches inspected; strata counts	Misaligned channels: assert shapes and CRS
M6	Baselines (D4–5)	`baselines.py` B0, B1	E1 numbers for B0/B1	B1 weight format: read repo loader

#	Milestone (days)	Tasks and files	Done when	Likely errors and fixes
M7	Training (D5–8)	`train.py`, `train_unet.yaml`, Colab notebook	Checkpoint + validation curves	Colab disconnects: checkpoint every epoch; overfit: early stop by event
M8	Evaluation and error analysis (D8–9)	`metrics.py`, E1, E2, E8, error gallery	Tables + 10 failure examples	Pooled metric hides strata: always stratify
M9	Core contribution (D6–9)	`terrain.py`, `change.py`, `gating.py`, optional `s2.py`	Gated masks on Trishuli	No threshold tuning on EMSR927; fix k=3 first
M10	Ablations and comparison (D9–11)	`ablations.py`, E3, E5	Ablation + same/cross-track tables	Seed noise: ≥3 seeds where affordable
M11	Geospatial inference (D6–11)	`exposure.py`, `network.py`, `flowpath.py`; build and test on synthetic hazards from D6	Statuses for Trishuli; E6	OSM topology gaps: split ways at shared vertices; unsnapped villages → "unknown"
M12	Backend and interface (D9–12)	`schema.py`, `streamlit_app.py` against mock `results.json` first	UI works on mock then real run	Heavy rasters in browser: serve PNG/COG overlays
M13	Integration (D12–13)	`pipeline.py`, caching, timing	One command runs two AOIs	Stage contract drift: schema tests
M14	Testing and reproducibility (D13–14)	Four tests, fresh-clone run, seeds, README	Teammate reproduces in a clean env	Hidden local paths: grep and fix

#	Milestone (days)	Tasks and files	Done when	Likely errors and fixes
M15	Deployment and demo (D14–15)	Rehearse live run on a non-Trishuli AOI; record video	Video ⩽3:00	Network failure: cached scenes
M16	Final documentation (D12–15)	Report ⩽6 pages, limitations, pitch, citations	Page count and checklist met	Overlong report: move tables to appendix files

Critical path: M3 → M4 → M7 → M8 → M10 → M13 → M15. M4 is the riskiest link. M11 and M12 run in parallel on mock data from day 6, so integration is not left to the end.

Pipeline sketches (verify endpoints before relying on them)

```
# OSM snapshot via ohsome (tile large bboxes)
curl -s -X POST https://api.ohsome.org/v1/elements/geometry \
  -d "bboxes=W,S,E,N" -d "time=2026-07-27" \
  -d "filter=highway=* and geometry:line" -d "properties=tags" >
roads.geojson
```

```
# Same-track selection (endpoint and field names to verify on CDSE)
from pystac_client import Client
cat = Client.open("https://stac.dataspace.copernicus.eu/v1")
items = list(cat.search(collections=["sentinel-1-grd"], bbox=aoi,
                        datetime="2026-07-15/2026-09-10").items())
# group by relative orbit + direction; need 1 post and >=1 pre at -12/-24/-36
d; else "no valid observation"
```

```
# Cut-off status under one closure scenario
d0 = nx.multi_source_dijkstra_path_length(G, dests, weight="t")      #
baseline
H = G.copy(); H.remove_edges_from(closed)                              #
scenario
d1 = nx.multi_source_dijkstra_path_length(H, dests, weight="t")
for s in settlements:
    if s not in d0:  status[s] = "unknown"        # no baseline route in OSM
    elif s not in d1: status[s] = "cut_off"
    elif d1[s] > 1.5 * d0[s]: status[s] = "degraded"   # 1.5x is a declared
choice
    else: status[s] = "connected"
```

Closure scenarios **[D]**: S-A conservative (any intersection with hazard probability ⩾ 0.5, 15 m tolerance for 10 m pixels and OSM positional error); S-B moderate (⩾50 m of road inside high-confidence hazard); S-C bridges only. The report shows the range S-C to S-A, not a single number.

15. Hardware, software, time and cost estimates

Item	Estimate	Basis
Laptop	16 GB RAM, ~150 GB free disk	SNAP is memory-hungry; S1 GRD scenes are large **[unmeasured]**
GPU	Free Colab T4, roughly 4–8 GPU hours total	My estimate for a ~10–15k patch subset; measure in M7
Inference	CPU, tiled	Small AOIs only; measure in E7
Software	All free/open source	Table in section 12
Cash	~$0 base; optional Colab Pro if sessions are cut	Quotas may change
Calendar	15 days; critical path ~11 days of serial work	Section 14

16. Risk register and fallbacks

Risk	L	I	Mitigation	Fallback preserving the PS
Kuro Siwo download too large/slow	M	H	Size check first; webdataset subset	Train on a smaller subset or use B1 only
No debris labels	H	M	Rule-based disturbance layer, clearly labelled	Water-only claim
CRS/coordinate mismatch	M	M	UTM zone from AOI centroid; assert CRS on every raster	Reproject at stage boundary

Risk	L	I	Mitigation	Fallback preserving the PS
Misregistration	M	M	Same-track only; check correlation on stable terrain	Phase-correlation shift
Class imbalance	H	M	Weighted loss, flood-biased sampling	Report per-class metrics
GPU limits	M	M	Subset, ResNet-18, fp16	B0 + B1 pipeline still delivers all PS questions
Poor steep-terrain performance	H	H	Gating; slope-stratified honesty	State the limit in the report
Domain shift to Himalaya	H	H	Adaptive thresholds; zero-tuning report	Present as measured transfer gap
Preprocessing too slow for live demo	H	H	Subset first; cache; precomputed Trishuli; fast mode	Show cached run plus live small AOI
CDSE outage or quota	M	H	Pre-downloaded scenes	Sentinel Hub fast path
No same-track post scene for judge AOI	M	H	Explicit "no valid observation" and next-pass date	Cross-track ablation, flagged not comparable
OSM gaps	H	M	"Unknown" status; dropout study	Report ranges
Integration incomplete	M	H	`results.json` contract and mock data from D6	Ship CLI + static report
Rule violation via leakage	L	Critical	Folder separation, leakage test, review	Remove the dependency before submission
Novelty claim fails prior-art check	M	L	Claims limited to engineering/evaluation	Reword

Three levels

- **Minimum viable:** same-track pairing, U-Net B3 + basic gating, exposure, one closure scenario with four statuses, Streamlit map, template sitrep, E1 + E4 + E6(i)(iii), tests, README, report, video.
- **Strong:** E2–E8, three scenarios, OSM dropout study, S2 corroboration, temperature scaling, bonus flow path, EMS closure replay.
- **Advanced (only if ahead):** Nepali template strings reviewed by a native speaker, conformal abstain class, a second Himalayan check scene (if reference polygons exist), SLC coherence.

17. Final demonstration scenario (3 minutes)

Time	Content
0:00–0:20	The question rescuers ask; one-line disclaimer (educational prototype); no victim imagery
0:20–1:20	Enter Trishuli AOI + 26 Aug: same-track pair shown, hazard layers appear, evidence/confidence toggle
1:20–2:00	EMSR927 comparison panel: agreement, disagreement reasons, stratified metrics, failure panel
2:00–2:30	Exposure counts and cut-off table with scenario toggle; "unknown" settlements explained
2:30–3:00	One-page sitrep, limitations in one sentence each, attributions

Rehearse a live run on a non-Trishuli AOI so the answer to "does it work anywhere?" is shown, not asserted.

18. Judging pitch

- **Problem:** after a debris flood, rescuers need to know where the valley changed and who lost road access, from satellites, quickly.
- **Existing gap:** flood masks, flooded-road overlays and healthcare-access analyses exist separately; Himalayan terrain false alarms and the link from hazard to isolation are rarely tested under held-out conditions.

- **Solution:** Cutoff runs same-track Sentinel-1 change detection with terrain gating and a trained model, then a scenario-based road-graph analysis.
- **Contribution:** engineering and evaluation, not a new architecture: automated pairing, gating, slope-stratified held-out tests, Trishuli check with zero tuning, cut-off sensitivity.
- **Evidence:** *(fill with measured E1, E2, E4, E6 numbers; none exist yet)*.
- **Impact:** a transparent, reproducible starting point; ranges and "unknown" instead of false certainty.
- **Limitations:** revisit gaps, 10 m pixels, smooth mud vs water, OSM gaps, exposure is not damage, educational prototype only.

19. References

Verified this session (**[V]**) unless marked.

- Official PS: Track_B.pdf / Space_Track.pdf (supplied).
- Kuro Siwo repo: https://github.com/Orion-AI-Lab/KuroSiwo [V]; paper: Bountos et al., NeurIPS 37 (2024), pp. 38105–38121, https://arxiv.org/abs/2311.12056 and https://proceedings.neurips.cc/paper_files/paper/2024/hash/43612b0662cb6a4986edf859fd6ebafe-Abstract-Datasets_and_Benchmarks_Track.html [V]; data: https://huggingface.co/datasets/orion-ai-lab/Kuro-Siwo-Webdataset and https://huggingface.co/datasets/orion-ai-lab/Kuro-Siwo-GeoTIFFs [V].
- Sen1Floods11: https://github.com/cloudtostreet/Sen1Floods11; doi 10.1109/CVPRW50498.2020.00113 [S].
- GEOID-Flood: https://arxiv.org/pdf/2608.02315; https://github.com/links-ads/geoid-flood [V].
- MMFlood: https://github.com/edornd/mmflood [V].
- EMSR927: https://mapping.emergency.copernicus.eu/news/flood-in-nepal-emsr927/ [V]; EU Space image pages: https://eu-space.europa.eu/components/earth-observation-copernicus/image-of-the-day/aftermath-nepal-flash-flood and https://eu-space.europa.eu/components/earth-observation-copernicus/image-of-the-day/copernicus-emergency-management-service-maps-flood-damages-northern-nepal [V].
- Event summary (secondary): https://gee-community-catalog.org/projects/nepal_floods_2026/ [V].

- ICIMOD 2017 Nepal flood extent: https://rds.icimod.org/metadata/8d97f509-a256-4e80-bf39-223b534a73fa [V]; NDRRMA 2025 sitrep: https://dpnet.org.np/resource-detail/2197 [V].
- ohsome API: https://docs.ohsome.org/ohsome-api/stable/ [V].
- Via Elicit/Docs A–C only **[S]**: Petricola et al. 2022 https://doi.org/10.1186/s12942-022-00315-2; Helleis et al. 2022 https://doi.org/10.1109/jstars.2022.3152127; Yang et al. 2021 https://doi.org/10.3390/rs13112205; github.com/GIScience/healthcare-access-idai; github.com/JakubNiedzwiedz/SAR-Flood-Detection; github.com/AliForghani/sentinel1-sar-flood-mapping.

Required attribution strings (copied from the PS; put them in the README, dashboard footer, report and sitrep):

```
"Contains modified Copernicus Sentinel data 2026."
"Produced using Copernicus WorldDEM-30 © DLR e.V. 2010–2014 and © Airbus
Defence and Space GmbH 2014–2018 provided under COPERNICUS by the European
Union and ESA; all rights reserved."
"© OpenStreetMap contributors."
```

Plus the Kuro Siwo citation and, for EMSR927 comparisons, "European Union, Copernicus Emergency Management Service data".

20. Final checklist

Verified in this session	Decided (design)	Needs experiment or confirmation
PS rules, weights, attribution strings	AI component = segmentation model	Kuro Siwo download size and event geography
Kuro Siwo: 43 events, 67,490 patches, 2 pre + 1 post, DEM/slope, 3 classes, MIT file vs CC BY text, trainer flags, released weights	Training on Kuro Siwo only	Contents of `grd_preprocessing.xml`; weight input format
GEOID-Flood and MMFlood are EMS-derived	Reject EMS-derived training sets	Same-track S1 availability and latency for the corridor

Verified in this session	Decided (design)	Needs experiment or confirmation
EMSR927 scope; four AOIs; >240 destroyed / 32 damaged in one AOI view	U-Net B3 + slope + gating; temperature scaling	EMSR927 vector layers and legend; whether EMS lists damaged roads/bridges
S2 scenes of 12 and 27 Aug exist near Betrawati/Gerkhu	Closure scenarios S-A/S-B/S-C; four statuses; 1.5× degraded rule	Every hypothesis H1–H6
Event timing and provisional-damage caveats from the community catalog	Scene-adaptive thresholds, k=3, HAND corridor start 60 m	pyflwdir API; Sentinel Hub quota; ohsome response limits
ohsome supports `time` and `filter`	Streamlit dashboard; template report, no LLM	Whether any second Himalayan check scene has reference polygons

Questions to put to the organisers before building around them

1. Will the live round allow pre-downloaded scenes and cached runs, and is there reliable internet for CDSE downloads?
2. Does Sentinel Hub (on CDSE) orthorectified Sentinel-1 count as "raw satellite data", or must preprocessing run locally?
3. Are high-resolution open images (Planet/Maxar, CC BY-NC) acceptable for visual checking only, or should they be avoided entirely?
4. Is a Nepali report expected under the segmentation option, or only under the copilot option?